



NORTH SYDNEY BOYS HIGH SCHOOL

2019 Year 11 Class Test

Mathematics Extension 1

General Instructions

- Working time – 60 minutes
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.
- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Hwang
- ☐ Dr Jomaa
- ☐ Ms Lee
- ☐ Mr Lin
- ☐ Ms Ziazaris
- ☐ Mr Zuber

Student Name: _____

(To be used by the exam markers only.)

Question No	1	2	3	4	5	Total	Total %
Mark	$\overline{14}$	$\overline{12}$	$\overline{9}$	$\overline{11}$	$\overline{8}$	$\overline{54}$	$\overline{100}$

Question 1 (13 marks)

(a) Factorise fully: $x^2 + 2x - y^2 - 2y$ 2

(b) Solve $|2x - 5| = 13$ 2

(c) Simplify fully: 3

$$\frac{4}{x^2 - 4} - \frac{3}{8 + 2x - x^2}$$

(d) Solve the following inequation: 3

$$\frac{4}{x - 6} \geq 3$$

(e) The fraction $\frac{\sqrt{15}+2}{\sqrt{3}+\sqrt{5}}$ can be written in the form of $a\sqrt{3} + b\sqrt{5}$. Find a and b 3

Question 2 (12 marks)

(a) Find the **domain** and **range** of the following functions:

(i) $f(x) = \sqrt{4 - x}$ 2

(ii) $f(x) = \frac{1}{2x+3}$ 2

(b) Sketch $f(x) = 2(x - 2)(x + 3)^2$ 3

(c) Given $f(x) = x^2 - 4x - 3$, evaluate $2f(x + a)$ 2

(d) Find the values of K for which the equation 3

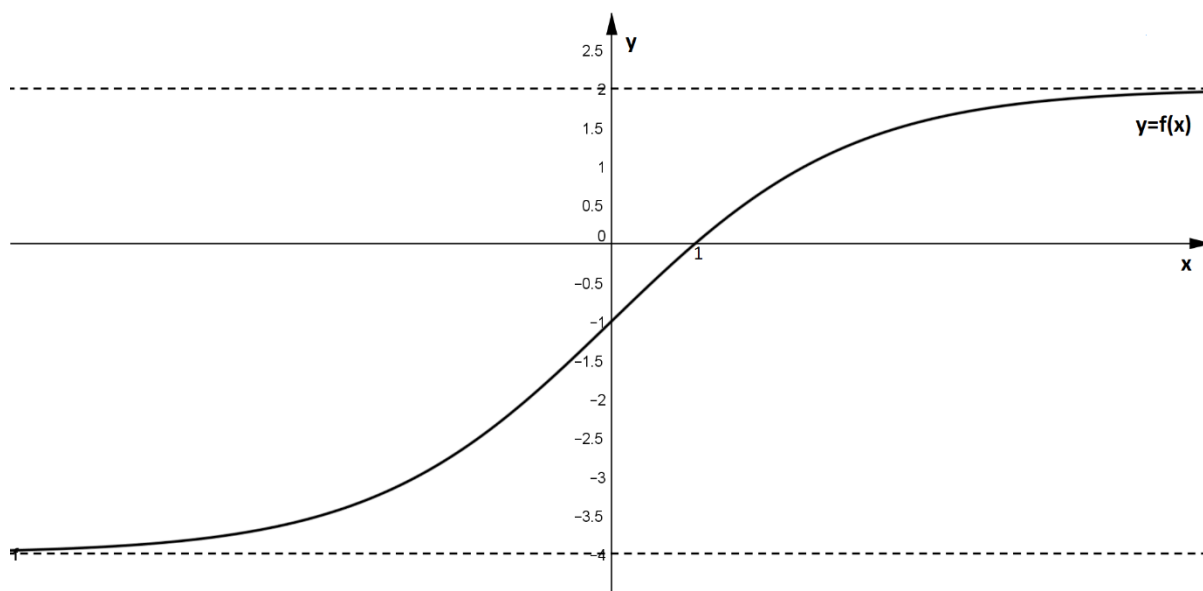
$$(2 - K)x^2 - 4(K - 2)x - 5 = 0$$

has two real distinct roots

Question 3 (9 marks)

- (a) Given the function $f(x) = \frac{x}{x^2+1}$. Determine if the function is ODD, EVEN or NEITHER. 2
 Showing all working.

- (b) Given the following function $y = f(x)$



Sketch the following graphs on the answer sheet provided:

- (a) $y = \frac{1}{f(x)}$ 2
 (b) $y^2 = f(x)$ 2
 (c) $|y| = f(-x)$ 3

Question 4 (11 marks)

- (a) Solve the trigonometric equation for $-180^\circ \leq \theta \leq 180^\circ$
- (i) $4\sin^2 x = 3$ 2
 (ii) $\sec 2x = \frac{2\sqrt{3}}{3}$ 3
 (iii) $2\sin^2 x = 5\cos x + 4$ 3

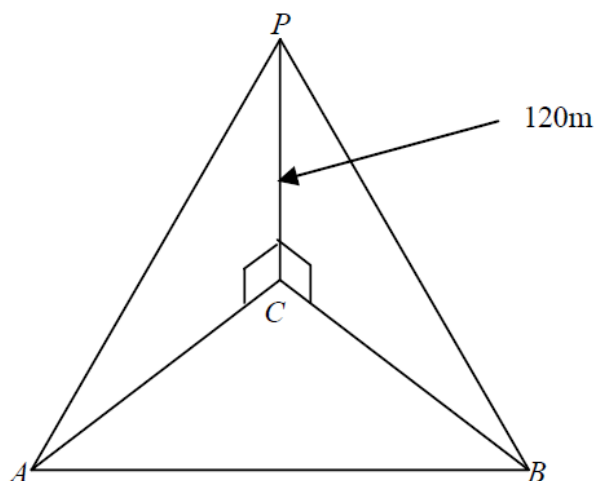
- (b) Simplify fully

$$\sqrt{\frac{\operatorname{cosec}^2 x - \cot^2 x - \cos^2 x}{\cos^2 x}} \quad \text{3}$$

Question 5 (8 marks)

(a)

Two boats at A and B are observed from the top P of a vertical cliff CP of height 120 metres. A is on a bearing of 195°T from the cliff and its angle of depression from P is 22° . B is on a bearing of 161°T from the cliff and its angle of depression from P is 27° .



- | | | |
|------|--|----------|
| (i) | Find $\angle ACB$ | 1 |
| (ii) | Find the distance between the boats (to the nearest metre) | 3 |

(b) The function $f(n)$ is defined on the set of integers and satisfies:

$$f(n) = \begin{cases} n - 3 & , n \geq 1000 \\ f(f(n + 5)) & , n < 1000 \end{cases}$$

- | | | |
|------|---------------------------|----------|
| (i) | Evaluate $f(999)$ | 2 |
| (ii) | Find the value of $f(84)$ | 2 |

End of Examination

Question 1

a) $|2x-5| < 13$

$$-13 < (2x-5) < 13$$

$$-8 < 2x < 18$$

$$-4 < x < 9$$

b) $x^2 + 2x - y^2 - 2y$

$$= x^2 - y^2 + 2x - 2y$$

$$= (x-y)(x+y) + 2(x-y)$$

$$= (x-y)(x+y+2)$$

c) $\frac{4}{(x-2)(x+2)} - \frac{3}{-(x-4)(x+2)}$

$$= \frac{4}{(x-2)(x+2)} + \frac{3}{(x-4)(x+2)}$$

$$= \frac{4(x-4) + 3(x-2)}{(x-2)(x+2)(x-4)}$$

$$= \frac{4x - 16 + 3x - 6}{(x-2)(x+2)(x-4)}$$

$$= \frac{7x - 22}{(x-2)(x+2)(x-4)}$$

d) $\frac{4}{x-6} \geq 3, \quad x \neq 6$

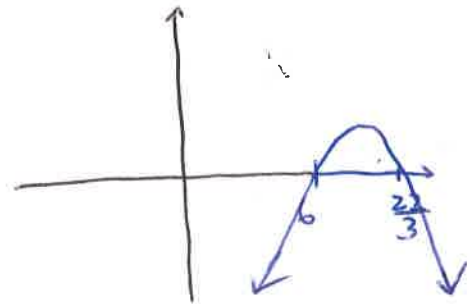
$$4(x-6) \geq 3(x-6)^2$$

$$4(x-6) - 3(x-6)^2 \geq 0$$

$$(x-6)[4 - 3(x-6)] \geq 0$$

$$(x-6)(4 - 3x + 18) \geq 0$$

$$(x-6)(22 - 3x) \geq 0$$



$$\therefore 6 < x \leq \frac{22}{3},$$

e) $\frac{\sqrt{15} + 2}{\sqrt{3} + \sqrt{5}} \times \frac{\sqrt{3} - \sqrt{5}}{\sqrt{3} - \sqrt{5}}$

$$= \frac{\sqrt{45} - \sqrt{75} + 2\sqrt{3} - 2\sqrt{5}}{-2}$$

$$= \frac{3\sqrt{5} - 5\sqrt{3} + 2\sqrt{3} - 2\sqrt{5}}{-2}$$

$$= \frac{\sqrt{5} - 3\sqrt{3}}{-2}$$

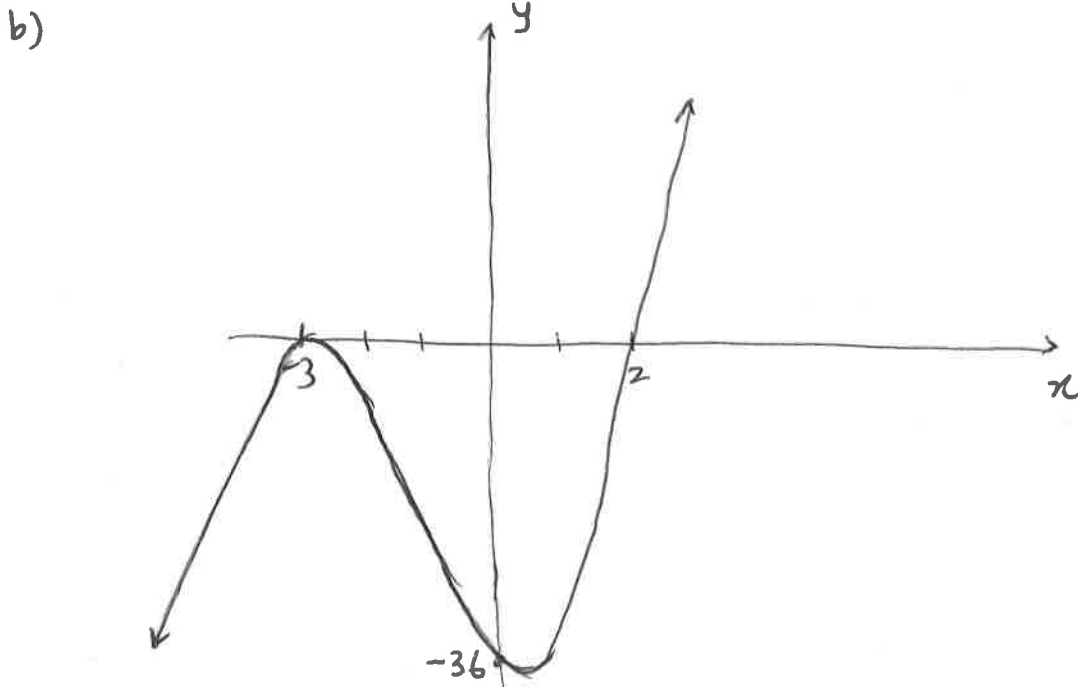
$$= \frac{3}{2}\sqrt{3} - \frac{1}{2}\sqrt{5}$$

$$\therefore a = \frac{3}{2} \quad b = -\frac{1}{2}$$

Question 2

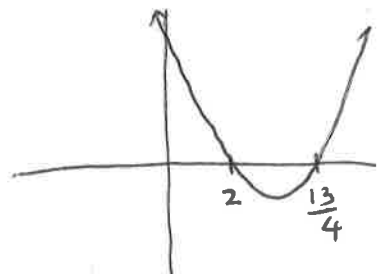
a) i) $D: x \leq 4$, $(-\infty, 4]$
 $R: y \geq 0$, $[0, +\infty)$

ii) $D: \text{all real } x, x \neq -\frac{3}{2}$
 $R: \text{all real } y, y \neq 0$



c) $f(x) = x^2 - 4x - 3$

$$\begin{aligned} 2f(x+a) &= 2((x+a)^2 - 4(x+a) - 3) \\ &= 2(x^2 + 2ax + a^2 - 4x - 4a - 3) \\ &= 2x^2 + 4ax + 2a^2 - 8x - 8a - 6 \end{aligned}$$



d) Two real distinct roots $\Delta > 0$

$$\Delta = b^2 - 4ac$$

$$\Delta > 0 \Rightarrow [-4(k-2)]^2 - 4(2-k)(-5) > 0$$

$$16(k-2)^2 + 20(2-k) > 0$$

$$4(k-2)[4(k-2)-5] > 0$$

$$4(k-2)(4k-13) > 0$$

$$k < 2 \text{ or } k > \frac{13}{4}$$

Question 3

a) $f(x) = \frac{x}{x^2+1}$

$$f(-x) = \frac{(-x)}{(-x)^2+1}$$

$$= -\frac{x}{x^2+1}$$

$$= -\left(\frac{x}{x^2+1}\right)$$

$$= -f(x)$$

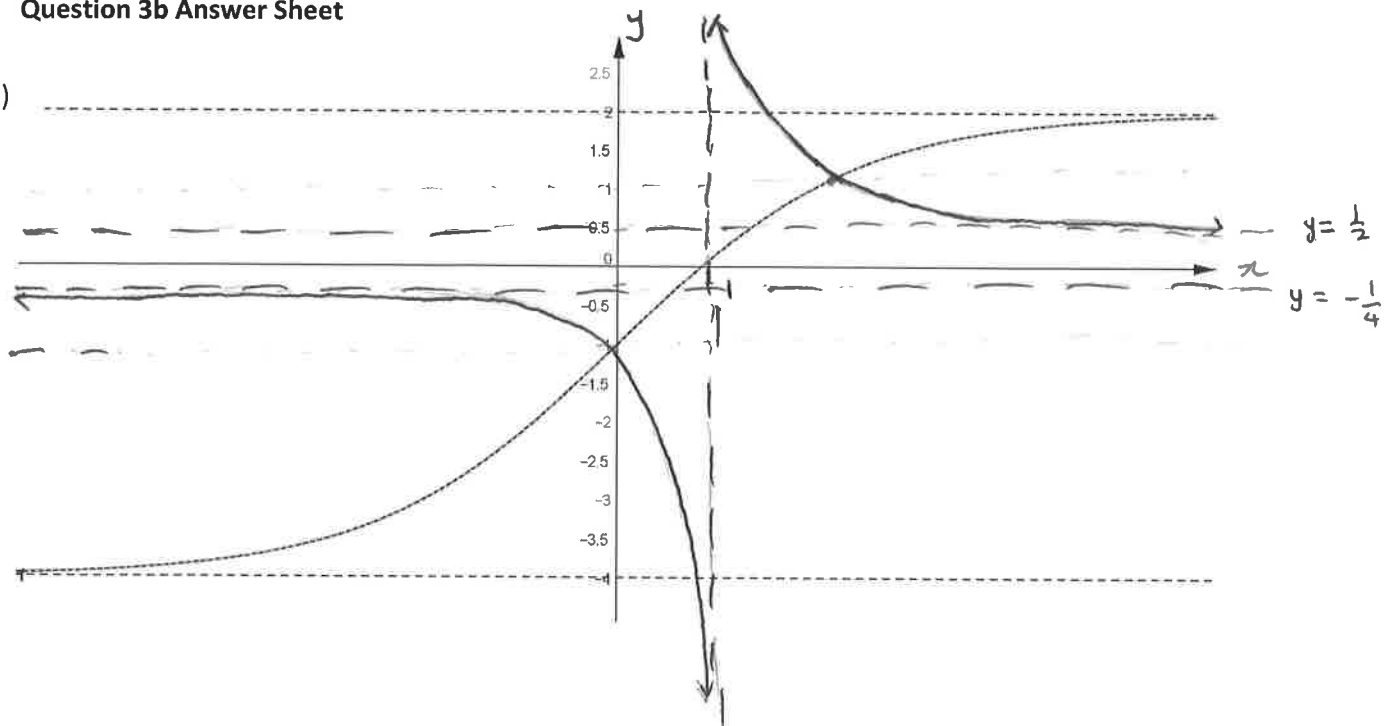
$\therefore f(x)$ is odd

b) See Answer Sheet

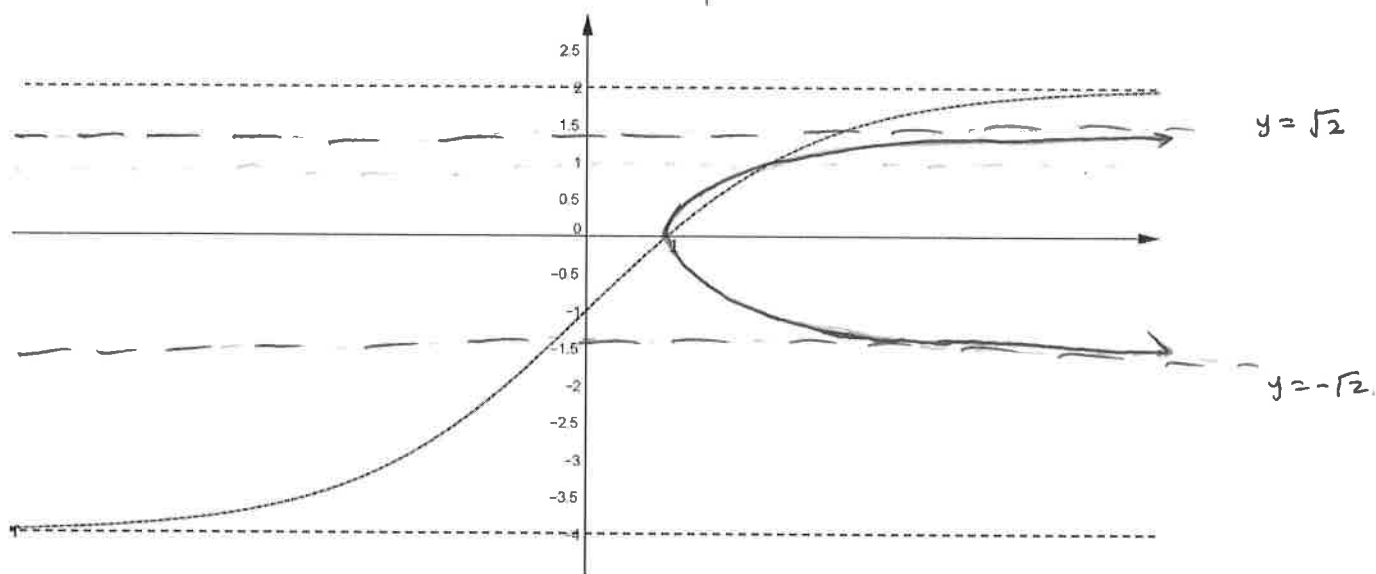
Question 3b Answer Sheet

Student Number

i)

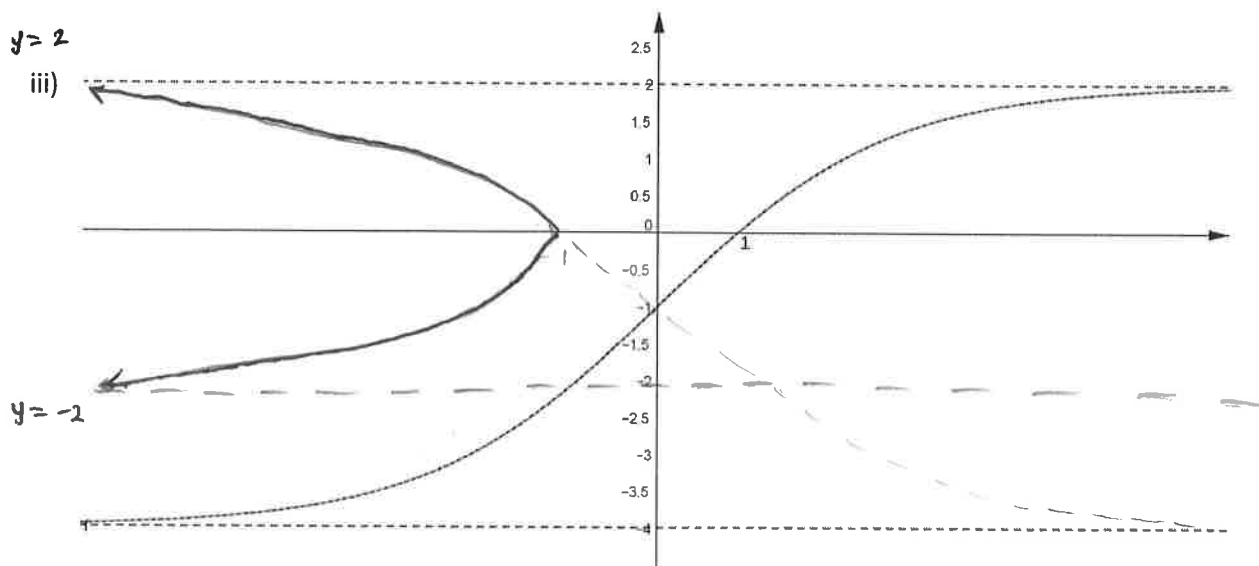


ii)



$y=2$

iii)



Question 4

a) $-180^\circ \leq x \leq 180^\circ$

i) $4\sin^2 x = 3$

$$\therefore \sin x = \pm \frac{\sqrt{3}}{2}$$

$$x = 60^\circ, 120^\circ, -120^\circ, -60^\circ$$

ii) $\sec 2x = \frac{2\sqrt{3}}{3} = \frac{2}{\sqrt{3}} \quad -360^\circ \leq 2x \leq 360^\circ$

$$\therefore \cos 2x = \frac{\sqrt{3}}{2}$$

$$2x = -30^\circ, 30^\circ, 330^\circ, -330^\circ$$

$$x = -15^\circ, 15^\circ, 165^\circ, -165^\circ$$

iii) $2\sin^2 x = 5\cos x + 4$

$$\therefore \sin^2 x = 1 - \cos^2 x$$

$$2(1 - \cos^2 x) = 5\cos x + 4$$

$$2\cos^2 x + 5\cos x + 2 = 0$$

$$(2\cos x + 1)(\cos x + 2) = 0$$

$$\cos x = -\frac{1}{2} \quad \text{or} \quad \cos x = -2 \quad (\text{which has no solutions})$$

$$x = -120^\circ \quad \text{or} \quad x = 120^\circ$$

b) $\sqrt{\frac{\operatorname{cosec}^2 x - \cot^2 x - \cos^2 x}{\cos^2 x}} = \sqrt{\frac{1 - \cos^2 x}{\cos^2 x}}$

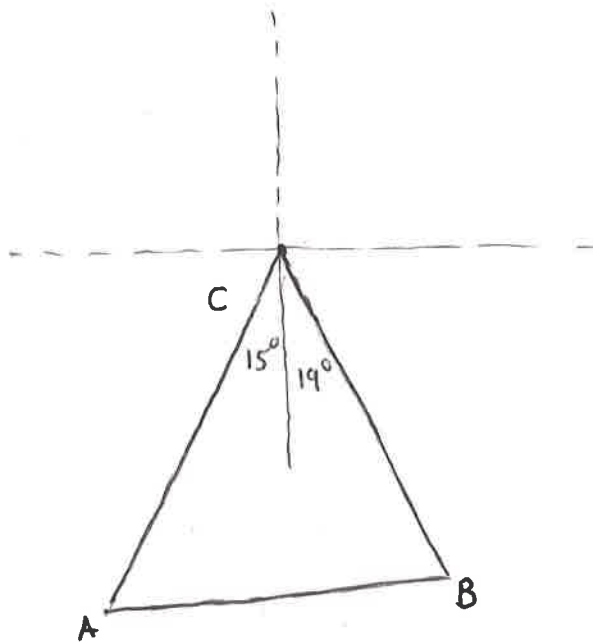
as $1 + \cot^2 x = \operatorname{cosec}^2 x$
 $\operatorname{cosec}^2 x - \cot^2 x = 1$

$$= \sqrt{\frac{\sin^2 x}{\cos^2 x}}$$

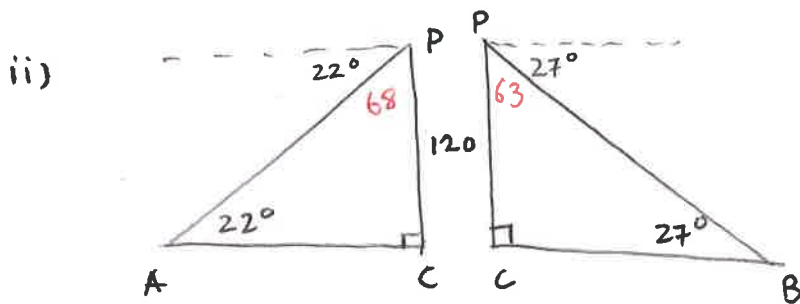
$$= \sqrt{\tan^2 x}$$

$$= |\tan x|$$

Question 5



i) $\angle ACB = 15^\circ + 19^\circ$
 $= 34^\circ$



$$\tan 22^\circ = \frac{120}{AC}$$

$$\therefore AC = 120 \cot 22^\circ \quad BC = 120 \cot 27^\circ$$

$$AB^2 = AC^2 + BC^2 - 2 \times AC \times BC \times \cos 34^\circ$$

$$= (120 \cot 22^\circ)^2 + (120 \cot 27^\circ)^2 - 2(120 \cot 27^\circ)(120 \cot 22^\circ) \cos 34^\circ$$

$$= 27699.51$$

$\therefore$

$$AB \approx 166 \quad (\text{to nearest metre})$$

b) i) $f(999) = f(f(1004))$
 $= f(1001)$
 $= 998$

ii) $f(998) = f(f(1003))$
 $= f(1000)$
 $= 997$

$$f(997) = f(f(1002))$$

$$= f(999)$$

$$= 998$$

$$f(996) = f(f(1001))$$

$$= f(998)$$

$$= 997$$

$$f(995) = f(f(1000))$$

$$= f(997)$$

$$= 998$$

function alternates.

$$f(84) = f(996)$$

$$= 997$$